

Local DNS Lab

By Keenan Morrison

First I created the debian template by installing the debian template to proxmox

```
The programs included with the Debian GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.  
root@pve1:~# ls  
root@pve1:~# pveam update  
update successful  
root@pve1:~# pveam available --section system | grep debian-13  
system      debian-13-standard_13.1-2_amd64.tar.zst  
root@pve1:~# pveam download  debian-13-standard_13.1-2_amd64.tar.zst  
400 not enough arguments  
pveam download <storage> <template>  
root@pve1:~# pveam download  local debian-13-standard_13.1-2_amd64.tar.zst  
downloading http://download.proxmox.com/images/system/debian-13-standard_13.1-2_amd64  
.tar.zst to /var/lib/vz/template/cache/debian-13-standard_13.1-2_amd64.tar.zst  
--2026-05-11 02:37:01--  http://download.proxmox.com/images/system/debian-13-standard  
_13.1-2_amd64.tar.zst  
Resolving download.proxmox.com (download.proxmox.com)... 66.70.154.82, 2607:5300:400:  
7d00::80
```

Figure 1.1

Then I added the hostname pi-hole and a complex password that I uploaded to a offline password manager

Create: LXC Container

General Template Disks CPU Memory Network DNS Confirm

Node: pve1 Resource Pool:

CT ID: 100 Password:

Hostname: pi-hole Confirm password:

Unprivileged container: ☐

Nesting: ☐

Add to HA: ☐

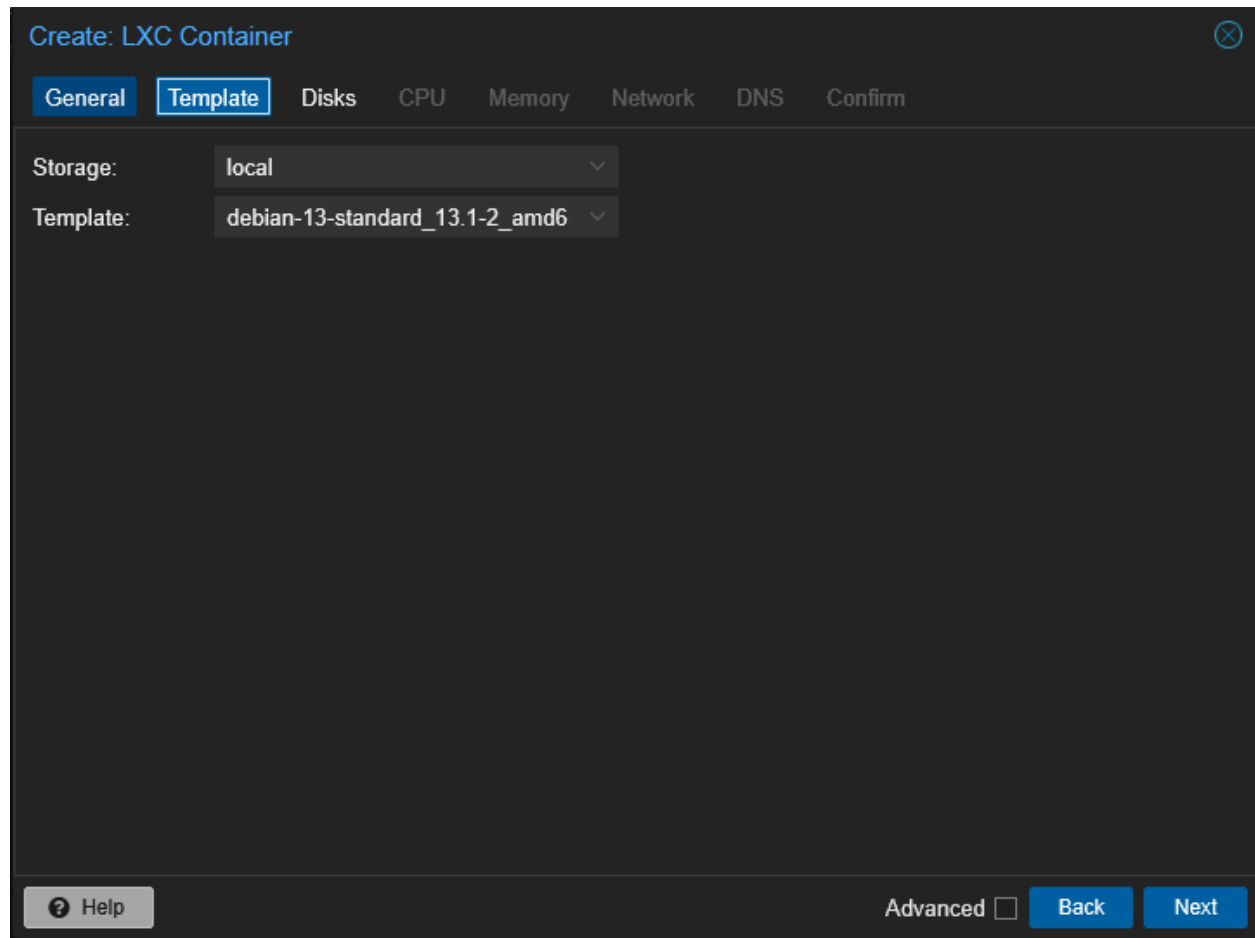
SSH public key(s):

Load SSH Key File

Help Advanced ☐ Back Next

Figure 1.2

I added the template Debian 13 that I created earlier



The screenshot shows a 'Create: LXC Container' window with a dark theme. At the top, there's a title bar with a close button. Below it is a tabbed interface with tabs for 'General', 'Template', 'Disks', 'CPU', 'Memory', 'Network', 'DNS', and 'Confirm'. The 'Template' tab is currently selected. In this tab, there are two dropdown menus: 'Storage:' set to 'local' and 'Template:' set to 'debian-13-standard_13.1-2_amd6'. At the bottom of the window, there is a 'Help' button with a question mark icon, an 'Advanced' checkbox which is currently unchecked, and 'Back' and 'Next' buttons.

Create: LXC Container

General Template Disks CPU Memory Network DNS Confirm

Storage: local

Template: debian-13-standard_13.1-2_amd6

? Help Advanced ☐ Back Next

Figure 1.3

Then I configured the CPU , Memory , Disks for the LXC container and it should look like this.

Create: LXC Container

General Template Disks CPU Memory Network DNS Confirm

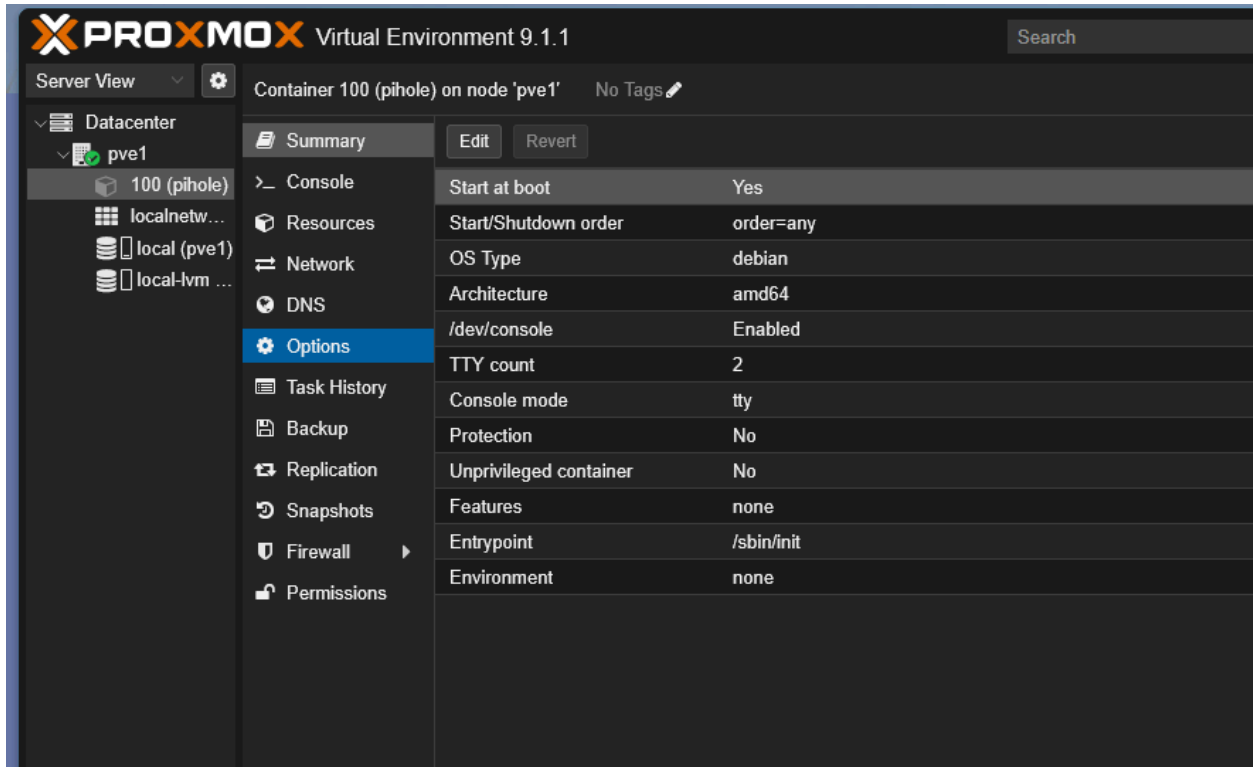
Key ↑	Value
cores	1
hostname	pihole
memory	512
net0	name=eth0,bridge=vbr0,firewall=1,ip=10.0.0.87/24,gw=10.0.0.1
nodename	pve1
osetemplate	local:vztmpl/debian-13-standard_13.1-2_amd64.tar.zst
pool	
rootfs	local-lvm:4
ssh-public-keys	
swap	512
unprivileged	0
vmid	100

☐ Start after created

Advanced ☐ Back Finish

Figure 1.4

The container ID is 100 as can be seen by this screenshot



The screenshot displays the Proxmox Virtual Environment 9.1.1 interface. The left sidebar shows the 'Datacenter' view with a tree structure including 'pve1' and its sub-items: '100 (pihole)', 'localnetw...', 'local (pve1)', and 'local-lvm ...'. The '100 (pihole)' container is selected. The main panel shows the 'Options' tab for this container, with a 'Summary' tab also visible. The 'Options' tab contains a table of configuration settings.

Property	Value
Start at boot	Yes
Start/Shutdown order	order=any
OS Type	debian
Architecture	amd64
/dev/console	Enabled
TTY count	2
Console mode	tty
Protection	No
Unprivileged container	No
Features	none
Entrypoint	/sbin/init
Environment	none

Figure 1.5

Then I went to pve1 and entered the command **pct enter 100**. After this I started the pihole script and this is what you should see.

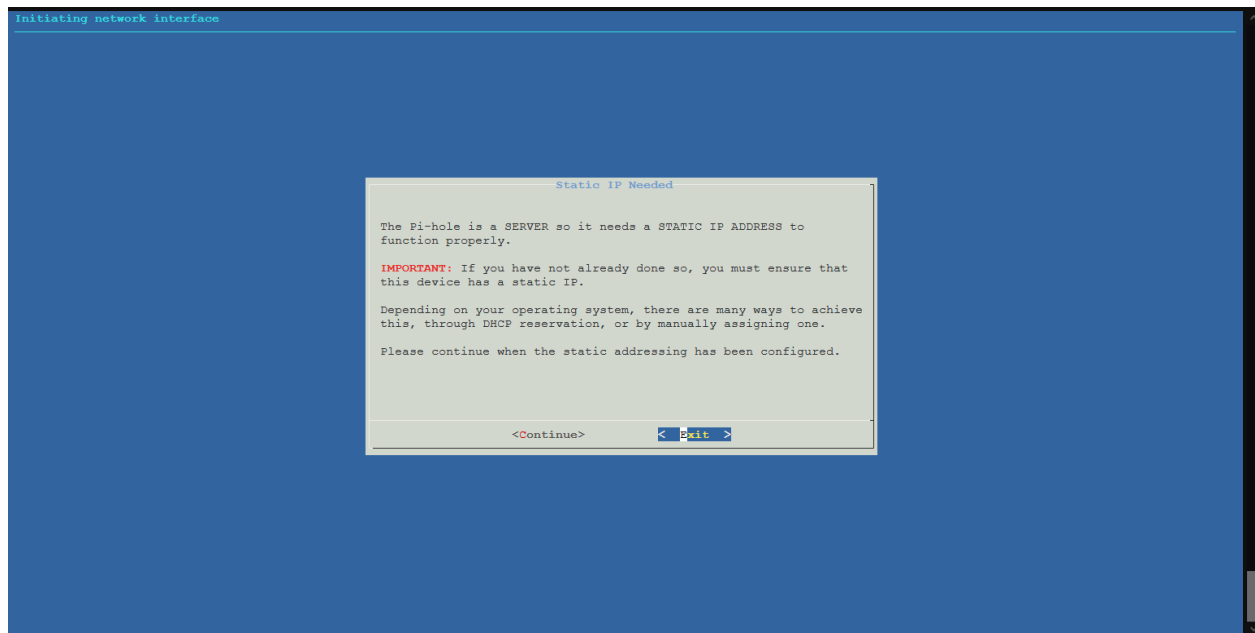


Figure 1.6

For the DNS I chose Cloudflare because I personally like it

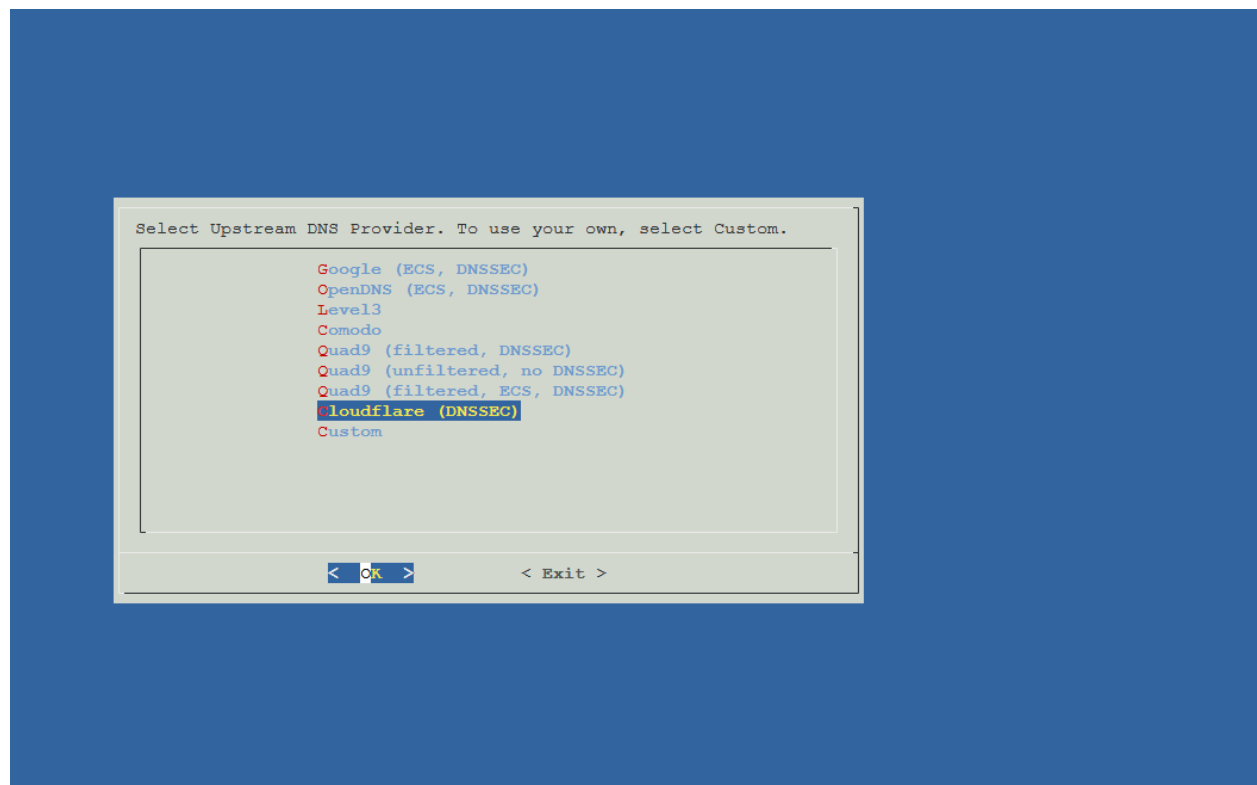


Figure 1.7

After that you should go to <http://IP ADDRESS/admin> you configure and you should see this dashboard

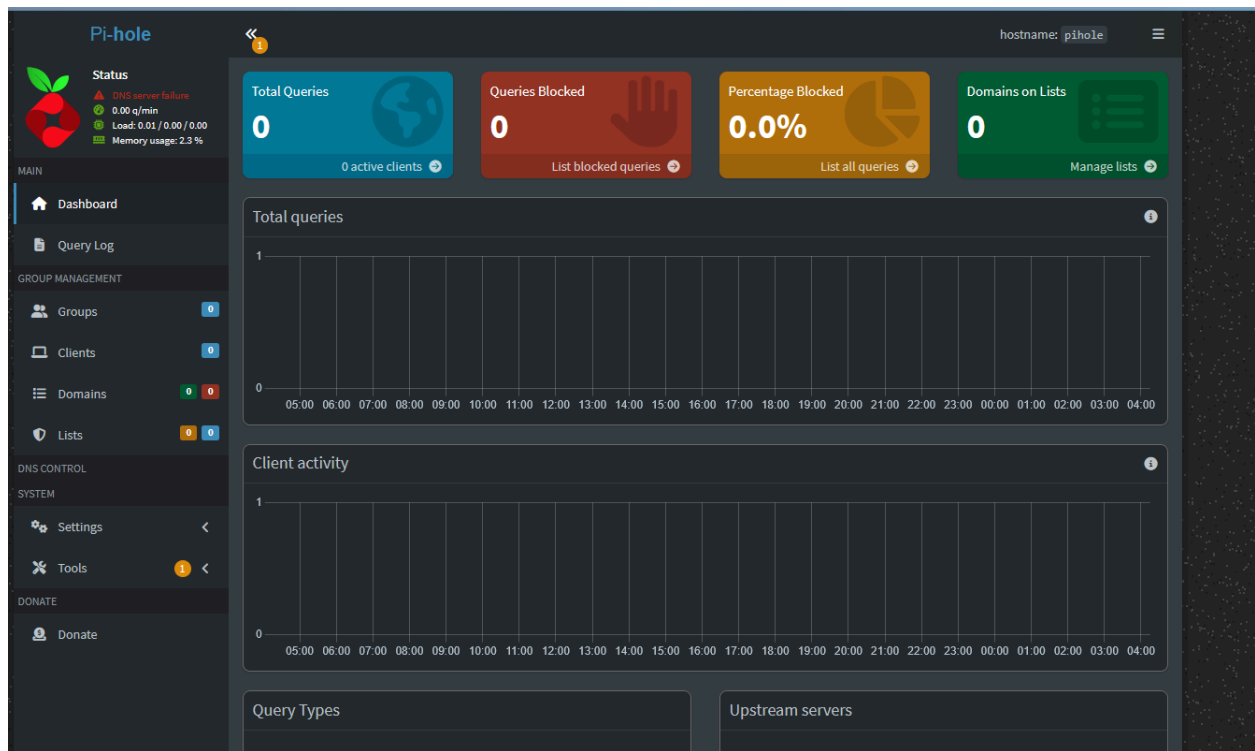


Figure 1.8

Then click Settings > Local DNS Records. It should look like the image shown in. When you are here then press the green plus button and add the ip of a locally hosted domain in the “Associated IP” section and for the domain put your desired domain name.

Pi-hole Status: Active, 50 q/min, Load: 0.17 / 0.15 / 0.07, Memory usage: 3.2 %

hostname: pihole

Local DNS Settings

Basic

Local DNS records

List of local DNS records

Show 10 entries Search: Previous 1 Next

Domain	IP	
linkwarden.local	192.168.0.191	
pihole.local	192.168.0.76	
pve1.local	192.168.0.78	
pve2.local	192.168.0.24	
router.local	192.168.0.1	
vaultwarden.local	192.168.0.188	
wazuh.local	192.168.0.110	
wiki.local	192.168.0.100	

Showing 1 to 8 of 8 entries

Domain Associated IP +

Local CNAME records

List of local CNAME records

Show 10 entries Search: Previous Next

Domain	Target	TTL *
No local CNAME records defined.		

Domain Target Domain +

* TTL in seconds (optional) Previous Next

Showing 0 to 0 of 0 entries

Note:
The target of a CNAME must be a domain that the Pi-hole already has in its cache or is authoritative for. This is a universal limitation of CNAME records.

The reason for this is that Pi-hole will not send additional queries upstream when serving CNAME replies. As consequence, if you set a target that isn't already known, the reply to the client may be incomplete. Pi-hole just returns the information it knows at the time of the query. This results in certain limitations for CNAME targets, for instance, only active DHCP leases work as targets - mere DHCP leases aren't sufficient as they aren't (yet) valid DNS records.

Additionally, you can't CNAME external domains (bing.com to

Figure 1.9

Overview

In this lab I set up a program called Pi-hole and configured it to be a locally hosted DNS server that can help me block ads and configure local DNS records to access services in my network a lot easier by committing words to memory instead of IPs.